

- 8 A crane is used to raise a weight of 200 N at a constant speed through a vertical height of 8.0 m in 4.0 s.

The efficiency of the crane is 20%. What is the electrical power needed to be supplied to the crane?

A 80 W

B 400 W

C 1600 W

D 2000 W

Ans: D

$$\text{Power to raise the weight} = \frac{200(8.0)}{4.0} = 400 \text{ W}$$

$$\text{Electrical power needed} = \frac{400}{0.2} = 2000 \text{ W}$$

- 9 Two discs P and Q of mass m and $2m$ respectively are placed on a rough horizontal and level